

Full Project: Movie Recommendation System

46

Import Libraries

- `import pandas as pd`
- `from sklearn.model_selection import train_test_split`
- `from sklearn.naive_bayes import GaussianNB`
- `from sklearn.metrics import accuracy_score, precision_score, recall_score`

47

Step 2: Load MovieLens Data

- `movies = pd.read_csv('movies.csv')`
- `ratings = pd.read_csv('ratings.csv')`
- `df = pd.merge(ratings, movies, on='movieId')`
- `print(df.head())`

48

Step 3: Preprocess Data

- `df['user_rating'] = df['rating'].apply(lambda x: 1 if x >= 4.0 else 0)`
- `df['genre_code'] = df['genres'].astype('category').cat.codes`
- `X = df[['genre_code']]`
- `y = df['user_rating']`

49

Step 4: Split the Data

- `X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)`

50

Step 5: Train Naïve Bayes Model

- `model = GaussianNB()`
- `model.fit(X_train, y_train)`

51

Step 6: Predict and Evaluate Model

- `y_pred = model.predict(X_test)`
- `print("Accuracy:", accuracy_score(y_test, y_pred))`
- `print("Precision:", precision_score(y_test, y_pred))`
- `print("Recall:", recall_score(y_test, y_pred))`

52

Step 7: Recommend Top Movies

- `like_probabilities = model.predict_proba(X)[: , 1]`
- `df['like_probability'] = like_probabilities`
- `top_movies = df[['title', 'genres', 'like_probability']].drop_duplicates().sort_values('like_probability', ascending=False)`
- `print("Top 10 Movie Recommendations:")`
- `print(top_movies.head(10))`

53

Build Web App Using Streamlit

- **Streamlit** is a **free, open-source** Python library that helps you **build web apps easily** especially for **data science, machine learning, and AI projects**.

54

Build Web App:

- Shows the **Top 10 movie recommendations**
- Displays **titles, genres, predicted like probability**
- **Shows posters** for movies if available!

55

Build Web App:

- pip install streamlit
- import streamlit as st

56

Build Web App:

- st.title("🎬 Smart Movie Recommender")
- st.write("Get your Top 10 movie recommendations based on real data!")
- if st.button("Show Top 10 Recommendations"):
- top_10 = top_movies.head(10)
- for index, row in top_10.iterrows():
- st.subheader(f"🎬 {row['title']}")
- st.write(f"Genre: {row['genres']}")
- st.write(f"👍 Likelihood you will like it: **{round(row['like_probability']*100, 2)}%**")

57

Build Web App:

- try:
 - `api_key = '549de4d0'`
 - `params = {'t': row['title'], 'apikey': api_key}`
 - `response = requests.get('http://www.omdbapi.com/', params=params)`
 - `data = response.json()`
 - `if data['Response'] == 'True' and data['Poster'] != 'N/A':`
 - `st.image(data['Poster'], width=200)`